

Artemis III: NASA's Orbital Dress Rehearsal for a Lunar Return

From: ScienceDaily Most Popular Mined: 2026-06-11 5 min read

NASA has unveiled the crew and ambitious plan for **Artemis III**, a mission that will push the boundaries of orbital operations like never before. Scheduled for 2027, it's a critical stepping stone toward Artemis IV, the first crewed lunar landing at the South Pole in 2028. The mission will see the **Orion spacecraft rendezvous** and dock with test versions of lunar landers from both Blue Origin and SpaceX a complex orbital ballet involving multiple launches of some of the most powerful rockets ever built.

The crew includes NASA astronauts Randy Bresnik, Frank Rubio, and Andre Douglas, along with ESA astronaut Luca Parmitano, marking the first time a European has been assigned to an Artemis mission. Bob Hines will serve as backup. "**Artemis III** will demonstrate the power of American innovation and international partnership," said NASA Administrator Jared Isaacman, calling it "the most awe-inspiring coordination of **heavy-lift** rocket launches in history."

Here's how it works: First, Blue Origin's lander pathfinder launches into orbit and waits. Then Orion lifts off atop the **SLS** rocket, carrying the crew. After system checks, Orion docks with the Blue Origin lander for about two days, during which astronauts will enter the lander and conduct tests. Orion then undocks and waits for SpaceX's Starship pathfinder to launch and dock for roughly a day of additional evaluations. Finally, Orion returns the crew to Earth with a Pacific Ocean **splashdown**.

The mission isn't just about docking. It will test software, communications, and propulsion systems, while also helping NASA increase launch cadence and strengthen supply chains for future deep-space missions, including eventual journeys to Mars. With hardware integration underway from Orion's docking system to **SLS** engine installations the path to a new Golden Age of exploration is taking shape. As ESA Director General Josef Aschbacher noted, "Europeans can take pride in being part of this exciting journey."

Word Treasure

Artemis III -- A NASA mission, planned for 2027, that will carry astronauts into lunar orbit to test docking with Moon landers before a later crewed landing.

Orion spacecraft -- The capsule designed to carry astronauts beyond low Earth orbit as part of the Artemis program.

rendezvous -- The carefully planned meeting and joining of two separate spacecraft while they are both traveling in orbit.

SLS -- The Space Launch System, NASA's most powerful rocket, built to send Orion and heavy payloads to the Moon.

heavy-lift -- Describes rockets capable of carrying very large and heavy cargo into space, like the SLS or SpaceX's Starship.

splashdown -- A return to Earth where the spacecraft lands in the ocean, a technique used since the Apollo era.

Background you need

The Artemis program is NASA's effort to return humans to the Moon for the first time since Apollo 17 in 1972, with the long-term goal of establishing a sustainable presence there. Artemis I, an uncrewed test flight of Orion and SLS, flew in 2022, and Artemis II, the first crewed flight around the Moon, is planned for 2025.

Artemis III is a critical middle step: instead of landing immediately, it will rehearse the orbital docking maneuvers needed for later missions. The landers are being built by two private companies: Blue Origin (founded by Jeff Bezos) and SpaceX (founded by Elon Musk). Their rivalry is fueling a new 'space race' in heavy-lift rocketry.

Jared Isaacman, the NASA Administrator, is known both as a billionaire entrepreneur who commanded private spaceflights and as a vocal advocate for commercial partnerships in space exploration. ESA Director General Josef Aschbacher oversees Europe's space activities and ensures that European astronauts and technology play a role in major international missions.

Luca Parmitano, an ESA astronaut from Italy, has already spent over a year in space and was the first Italian to walk outside the International Space Station. His assignment to an Artemis mission signals a deepening cooperation between NASA and Europe in deep-space exploration.

Break it down

LEAD: NASA announces crew and plan for Artemis III -- an orbital rendezvous rehearsal before a lunar landing.

+ detail: 'Scheduled for 2027, stepping stone toward Artemis IV in 2028'

KEY TECHNICAL CHALLENGE: A complex orbital ballet -- multiple launches, docking with two different lander prototypes.

+ HOW: Orion docks first with Blue Origin's lander pathfinder, then later with SpaceX's Starship pathfinder.

+ ROCKETS: SLS for Orion, separate rockets for each lander.

CREW: International lineup -- three NASA astronauts (Bresnik, Rubio, Douglas) plus ESA's Luca Parmitano.

+ backup: Bob Hines

ADMINISTRATOR QUOTE: Jared Isaacman calls it 'the most awe-inspiring coordination of heavy-lift rocket launches in history.'

MISSION SEQUENCE: Step-by-step breakdown -- Blue Origin lander launch, Orion launch, docking tests, undocking, Starship pathfinder launch, second dock, final splashdown.

BROADER GOALS: Beyond docking, the mission will test software, communications, propulsion, and help build up launch cadence and supply chains for future Mars missions.

CLOSING NOTE: ESA Director General Aschbacher emphasizes European pride and partnership.

Why it matters

This mission isn't just about a single crew; it's about proving that the complex orbital

choreography needed for a permanent Moon base -- and eventually Mars trips -- works in practice. For a 12-year-old, it means the era of routine deep-space voyages is getting closer, and with it, careers in engineering, science, and even living on other worlds.

Test what you remember

1. What is the main purpose of the Artemis III mission described in the article?
 - (A) Landing astronauts on the Moon's South Pole
 - (B) Testing orbital docking with lunar lander pathfinders
 - (C) Building a space station around Mars
 - (D) Sending a rover to search for water
2. Which two companies are providing test landers for the Artemis III mission?
 - (A) Boeing and Lockheed Martin
 - (B) Blue Origin and SpaceX
 - (C) Northrop Grumman and Virgin Galactic
 - (D) ESA and JAXA
3. Which European astronaut will fly on Artemis III, marking a first for ESA?
 - (A) Luca Parmitano
 - (B) Andre Douglas
 - (C) Bob Hines
 - (D) Frank Rubio
4. How does Orion reach orbit for this mission?
 - (A) On top of a SpaceX Falcon Heavy
 - (B) By launching on the SLS rocket
 - (C) Inside a Blue Origin New Glenn
 - (D) Using a Russian Soyuz rocket
5. After the orbital tests, where will Orion's crew land back on Earth?
 - (A) In the Atlantic Ocean
 - (B) On a runway in Florida
 - (C) In the Pacific Ocean
 - (D) In a desert in Kazakhstan
6. Beyond the Moon, the article suggests that technologies tested in Artemis III could eventually support journeys to which destination?
 - (A) Jupiter's moons
 - (B) Venus
 - (C) Asteroid belt
 - (D) Mars

Answer key (try the quiz first!): 1: B 2: B 3: A 4: B 5: C 6: D

